Question	Answers	Extra information	Mark	AO / Spec Ref.
03.9	as light (intensity) increases, rate / photosynthesis increases (rate / photosynthesis increases) at the same rate as light (intensity)	allow positive correlation allow rate / photosynthesis increases linearly / evenly allow description of shape of line eg straight / linear allow rate is (directly) proportional to light (intensity) for **2** marks	1 1	AO3 4.4.1.2

Total Question 3	**15**

Question 4

Question	Answers	Extra information	Mark	AO / Spec Ref.
04.1	vector		1	AO1 4.3.1.5

Question	Answers	Extra information	Mark	AO / Spec Ref.
04.2	any **two** from: • (it has) mitochondria • (it has a) nucleus • (it has) no plasmids	if neither mark awarded, allow **1** mark for protist has membrane-bound structures allow it does not have a loop of DNA allow other appropriate features do **not** accept (it has) cytoplasm do **not** accept (it has) a (cell) membrane do **not** accept it has no cell wall	2	AO2 4.3.1.5 4.1.1.1

Question	Answers	Extra information	Mark	AO / Spec Ref.
04.3	Salmonella		1	AO2 4.1.1.1 4.3.1.3

Question	Answers	Extra information	Mark	AO / Spec Ref.
04.4	only one parent is involved		1	AO1 4.3.1.5 4.6.1.1

Question	Answers	Extra information	Mark	AO / Spec Ref.
04.5	14		1	AO2 4.3.1.5 4.1.2.2

Question	Answers	Extra information	Mark	AO / Spec Ref.
04.6	decreased haemoglobin		1	AO2 4.3.1.5 4.2.2.3

Question	Answers	Extra information	Mark	AO / Spec Ref.
04.7		allow alternative route allow correct conversion to standard form at any stage		AO2 4.3.1.5
	$\frac{210\ 000\ 000}{2}$ **or** 105 000 000	allow $\frac{210 \text{ million}}{2}$ **or** 105 million	1	
	105 000 000 × 3	allow 105 million × 3	1	
	315 000 000	allow 315 million	1	
	3.15×10^8		1	

Question	Answers	Extra information	Mark	AO / Spec Ref.
04.8		ignore vaccination / anti-malarial drugs		AO2 4.3.1.5
	any **two** from:		2	
	• wear long(er) sleeves / clothes	allow reduce bare skin exposure		
	• use insect repellent or insecticides	allow insect repellent methods such as citronella candles		
	• don't go out in the evening	allow close doors / windows in the evening		
	• avoid going to countries / places with malaria / mosquitos			
	• destroy breeding grounds	allow methods of destroying breeding grounds such as drain water holes		
	• release sterile male mosquitos			
		allow other correct suggestions		

Question	Answers	Extra information	Mark	AO / Spec Ref.
04.9	percentage / chance (of getting malaria) decreases (with age)	do **not** accept percentage / chance of having disorder **S** decreases with age	1	AO3 4.2.2.5 4.3.1.5
	correct use of data such as two pairs of data from **Table 4**		1	
		allow use of processed data		

Total Question 4	**15**

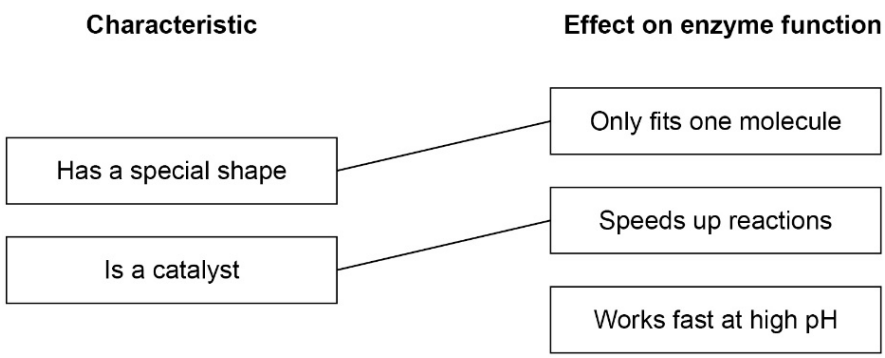
Question 5

Question	Answers	Extra information	Mark	AO / Spec Ref.
05.1	any **one** from: • growth • development	allow replacement (of old cells) allow for repair allow to heal wounds	1	AO1 4.1.2.2

Question	Answers	Extra information	Mark	AO / Spec Ref.
05.2	amino acids		1	AO1 4.2.2.1

Question	Answers	Extra information	Mark	AO / Spec Ref.
05.3	Biuret reagent		1	AO1 4.2.2.1 RPA4

Question	Answers	Extra information	Mark	AO / Spec Ref.
05.4	purple		1	AO1 4.2.2.1 RPA4

Question	Answers	Mark	AO / Spec Ref.
05.5	 do **not** accept more than one line from a box on the left	1 1	AO1 4.2.2.1